

Bridgetek. These are meant for use in retail, digital signage and office management systems, as well as public information units, vending machines, domestic appliances, store/mall direction finding systems,



training/educational equipment, and interactive exhibits. Each has 1MB embedded graphics RAM, which can be supplemented by attaching an external NOR flash memory through the QSPI interface. BT818 is optimised for use with 4-wire resistive touchscreens, while BT817 for capacitive touchscreens (with detection of up to 5 different touch-points).

Bridgetek
www.brtchip.com

DC-DC converter

For use in convection-cooled equipment and applications, including microcon-



trollers, data-com, telecom, remote sensor systems, and portable electronics, a DC-DC converter series has been introduced by CUI. Its features include 2:1 and 4:1 input voltage range, 3W power, single regulated output, 1,500V DC isolation, 125 to 600mA output current, -40 to +85 degrees Celsius operating temperature, low profile, and protection against over-voltage, over-current, and short-circuit.

CUI Inc.
www.cui.com

GNSS receiver

The new u-blox M10 is a highly integrated global navigation satellite system (GNSS) designed for ultra-low power, high-performance positioning applications such as sports watches or asset and livestock trackers. It can track up to four GNSS constellations at once to deliver positioning data even in challenging environments. The receiver's Super-S technology helps distinguish positioning signals from background noise to capture positioning data even when satellite signals are



weak. The device consumes 12mW in continuous tracking mode, making it ideal for battery-powered applications.

u-blox

www.u-blox.com

SENSORS

Digital magnetic sensor

Coto Technology has released RedRock RR122-1E73-511/512 wafer-based tunneling magnetoresistance (TMR) digital omnipolar sensor, featuring an operating frequency of 2,500Hz, a voltage supply range of 1.7V to 5.5V, and an extremely high magnetic sensitivity (15G or 1.5mT) in an ultra-miniature package at a competitive price. The sensor can be used in products ranging from small battery-powered electronics to industrial machinery. Specific applications could include proximity detection, rotary sensing, utility meters, portable rechargeable medical devices, motor controllers, and consumer electronics.



Coto Technology
www.cotorelay.com

Health sensors

The Health Sensor Platform 3.0 (HSP 3.0) from Maxim monitors blood oxygen saturation (SpO2), electrocardiogram (ECG), heart rate (HR), body temperature, and motion. Included algorithms provide HR, heart-rate variability (HRV), respiration rate (RR), SpO2, body temperature, sleep quality and stress level information at clinical-grade levels. It includes:



MAX86176. For synchronous acquisition of PPG and ECG measurements, providing pulse transit time for cardiac health.

MAX20360. Having integrated power and battery management power management IC optimised for advanced body-worn health sensing devices.

MAX32666. Bluetooth (BLE)-enabled, ultra-low-power microcontroller with two Arm Cortex-M4F cores and an additional SmartDMA, which permits running the BLE stack independently.

MAX32670. Ultra-low-power microcontroller for PPG algorithms of pulse rate, SpO2, HRV, RR, sleep quality and stress monitoring.

MAX30208. Reads the temperature and can be mounted on a flex cable or PCB, making it easier to incorporate into wearables.

Maxim Integrated

www.maximintegrated.com

CMOS image sensor

The new XGS family of image sensors from ON Semiconductor include the XGS



45000, XGS 30000, XGS 20000, and XGS 5000. These provide 12-bit image quality at a high frame rate,

detailed imaging with up to 45-megapixel resolution, and up to 60fps in 8K video mode. These are meant for challenging IoT applications such as machine vision and intelligent transportation systems. A global shutter ensures that moving objects can be captured without any motion artefacts.

ON Semiconductor

www.onsemi.com

ENTERPRISE-CLASS PRODUCTS

High-performance computer

For high-performance connectivity, the DA-820C series fanless 48.3cm (19-inch) 3U rackmount computer from Moxa features a powerful 7th generation Intel Xeon or Core i7/i5/i3 processor and multiple expansion interfaces to include IRIG-B, serial, and Gigabit Ethernet fibre SFP connectivity for smart grid applications. By using only one high-performance computer to run multiple operating systems on virtual machines, instead of installing multiple computers to run each OS separately, saves space and costs. To handle high-speed, low-latency communications for exchanging large amounts of data, while withstanding high EMI/EMC interference, the DA-820C features SFP interface for gigabit Ethernet fibre-optic communications.



Moxa
www.moxa.com